Java Programming

CPT111 – Lab 11

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CPT111 Java Programming

Lab 11

Recursion

Lab Exercise #11.1

- In this lab, you will learn problem solving using **recursion** and **recursion with helper methods**

- Complete the **helper method `smallest`** that takes an integer array and an integer

Complete the helper method `smallest`.

It finds the smallest integer in an integer array *recursively*.

You may assume that input array has at least one element in it

You must **not** use any loops or regular expressions.

Test cases:
`smallest`([10, 5, 7, 9]) → 5

Lab Exercise #11.1 – Activity 1: Base Case

- The helper method takes two parameters: the original array, and an integer start to mark the beginning of the ***remaining array*** that has not been considered
- Follow the Recursion Recipe in Lecture 11, to **find the base case**
  - Hint: the simplest case is when the *remaining array* is just one element

```java
// Lab Exercise #12.1 Recursive Smallest Integer
// Appear in FINAL EXAM 2019
public static int smallest(int[] array) {
    return smallest(array, 0);
}

private static int smallest(int[] array, int start) {

    // base case



    // recursive step



}
```

Lab Exercise #11.1 – Activity 2: Recursive Step

- Follow the Recursion Recipe in Lecture 11, to **find the recursive step**
  - Try some small integer array with some value of start
  - Call the function recursively on start + 1
  - How to *compute* the smallest integer given the answer to the recursive call with start + 1, which is *a smaller instance*?

```
7  private static int smallest(int[] array, int start) {
8
9      // base case
10
11
12
13     // recursive step
14
15
16
17 }
```

Lab Exercise #11.1 – Activity 3: Base Case is Always Reached

- Follow the Recursion Recipe in Lecture 11, to **argue that the base case is always reached**
  - we need to ensure that there will **not** be an *infinite recursion*!

```java
// Lab Exercise #12.1 Recursive Smallest Integer
// Appear in FINAL EXAM 2019
public static int smallest(int[] array) {
    return smallest(array, 0);
}

private static int smallest(int[] array, int start) {

    // base case



    // recursive step



}
```

Lab 11 Challenge – Recursive Replace Odd with Zero

- In the lab challenge, you will repeat the activities in the lab exercise with a similar problem using **a list**, instead of an array:

- Complete the **helper method** that takes an integer list **`list`** and an integer **`start`**:

Complete the helper method `replaceOddZeroHelper`.

It replaces *all* odd numbers with zero in the input integer list *recursively*.

You must **not** use any loops or regular expressions.

The imported libraries are Arrays and List.

Test cases:
`replaceOddZero`([1, 2, 3, 4, 5]) → [0, 2, 0, 4, 0]

For example:

Test	Result

`List<Integer> list = Arrays.asList(1, 2, 3, 4, 5);`
`replaceOddZero(list);`
`System.out.println(list);`
	`[0, 2, 0, 4, 0]`

Lab 11 Challenge with AI Agents

- You must use our **CPT111 AI Agents** to complete the Lab 11 Challenge

CPT111 AI Agents

CPT111_AI_helper

to do **3 things:**

1. **Generate test cases** to better evaluate your code
   "Give me five more test cases for Lab 11 Challenge to test my code"
2. **Identify bugs** in your code
   Copy paste your code and *"Check my code for Lab 11 Challenge for bugs and explain"*
3. **Generate 1-2 related questions**, for you to answer afterwards in the chat,
   so then TAs can read and give grade on your understanding of Lab 11 materials

- You can ask follow-up questions, to better your understanding

Lab 11 Challenge – Task 1: Base Case (20 marks)

- Follow the Recursion Recipe in Lecture 11, to **find the base case**
- What is the smallest instance?
  - what operations are needed, if any?
  - what should be returned?

```java
// Lab 12 Challenge Problem
public static void replaceOddZero(List<Integer> list) {
    replaceOddZeroHelper(list, 0);
}

private static void replaceOddZeroHelper(List<Integer> list, int start) {
    // Base case


    // Recursive step


}
```

Lab 11 Challenge – Task 2: Recursive Step (40 marks)

- Follow the Recursion Recipe in Lecture 11, to **find the recursive step**
  - show to TA how some examples relate to their smaller instances
  - given a recursive call the smaller instance, what operations are needed?

```java
// Lab 12 Challenge Problem
public static void replaceOddZero(List<Integer> list) {
    replaceOddZeroHelper(list, 0);
}

private static void replaceOddZeroHelper(List<Integer> list, int start) {
    // Base case


    // Recursive step


}
```

Lab 11 Challenge – Task 3: Base Case Reached and Solve (40 marks)

1. **Argue** that the **base case** is **<u>always reached</u>**!
2. **Show** to your Lab TA that you have passed the Learning Mall Quiz:

	Test	Expected	Got	

✔	`List<Integer> list = Arrays.asList(1, 2, 3, 4, 5);`
	`replaceOddZero(list);`
	`System.out.println(list);`
		`[0, 2, 0, 4, 0]`	`[0, 2, 0, 4, 0]`	✔

Passed all tests! ✔

3. **Use CPT111 AI Agents** to generate **1-2 questions** related to your work (Task 1-3) so far, and then answer them; for example:
   " I have completed the Lab 11 challenge question, and I am now ready to answer a question related to my solution. Could you please provide a question to check my understanding? "
4. **Show the TA your chat conversation**, for CW2 Grading

Thank you for your effort!

- **Please fill out a short questionnaire**
  - Once you have completed it, you will receive **a reward** as our thanks: access to **a past exam question with its solution**, as well as future access to additional ***past exam questions quiz*** for practice!

- You may leave the lab once you have complete the Lab 11 Challenge and it has been ASSESSED by a TA!
  - Please note that the **Lab 11 Challenge** is **graded** and contributes to your CW2 marks